Hardness Evidence Lab

I can use the same stated scratch method to compare four coded rock samples and give a cautious evidence-based conclusion.

LOCK
fair method

KEEP
no cherry-picking

CONCLUDE
name a limit

[Open the original interactive lesson]

What success looks like

- I can record an after-wipe result for all four codes.
- I can keep every valid result, including an unexpected one.
- I can compare relative scratch resistance and state one method or sample limitation.

Retrieve and reconnect

- Define relative hardness using scratch resistance.
- Name three controls in the stated method.
- Explain why the first pale line is not yet the final observation.
- Lead-in: state why every valid result must stay in the table.

Codes A–D give four different after-wipe results. Which recording rule protects the evidence?

Think first. Point, say, sign or write your choice. Give one reason.

A. Keep every valid result and note any safety stop or repeat.

B. Keep only the result that matches the prediction.

C. Choose harder until all four grooves are equally deep.

Look closely · what is the evidence?

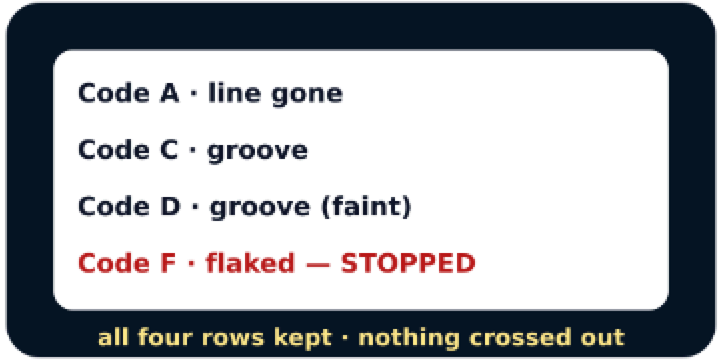
CODE A · deep groove remains

CODE B · shallow groove remains; loose grain triggers a check

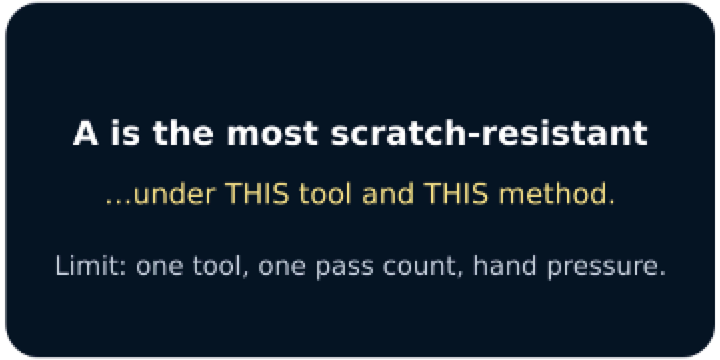
CODES C–D · faint groove / no groove after wiping

Look closely · what is the evidence?

The locked method: same tool, guide and passes



A results table with all four codes recorded



A cautious conclusion with its limit

Words we will use

- relative — compared with the others
- repeat — same method again
- conclusion — what the results support

Watch the model

- Lock the method before Code A: same secured tool, dry start, guide length, passes, safe force proxy, wipe and viewing rule.
- Predict, run or reveal, wipe and record one code at a time. Keep the table visible and never alter an earlier result.
- If the method fails or the sample flakes, write the stop note, repair access or setup safely and use a pretested parity result.

Code C gives a faint groove when you predicted no groove. The method was followed safely. What should you do?

A. Keep the faint-groove result and, if useful, repeat the same method.

B. Delete it because it does not match the prediction.

C. Choose harder until the groove looks like the expected result.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Order only the tested codes by the remaining groove evidence: deeper groove means lower resistance to this scratch method.
- Use a cautious conclusion such as ‘Code D resisted a groove more than Code A under this method.’
- Add one limitation: specimens vary, the force proxy is simple and one classroom tool does not produce a universal hardness number.

Find and repair the misconception

Evidence glitch: keep only the deepest and no-groove results because the middle results make the pattern untidy.

A. Keep it: tidy evidence is better than complete evidence.

B. Repair it: keep every valid result so the full relative pattern and variation remain visible.

C. Repair it: invent one average groove and erase the raw observations.

Four-Code Hardness Evidence Matrix

Place each after-wipe result, method card and limitation in its correct evidence role.

- Lock the method.
- Predict one code.
- Record the after-wipe result.
- Keep or repeat without erasing.
- Compare and limit the conclusion.

Four-Code Hardness Evidence Matrix

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- DEEP GROOVE
- SHALLOW GROOVE
- FAINT GROOVE
- NO GROOVE AFTER WIPE
- METHOD / REPEAT RULE
- LIMIT OF THE CONCLUSION

State the relative pattern, quote two codes and name one result-handling rule and one limitation.

Your independent mission

Run or direct the same safe scratch method on four coded samples, record every after-wipe result, repeat one check where appropriate and write a cautious conclusion.

Record: A complete four-code raw-results table, one preserved repeat/check note, a relative comparison using at least two codes and one method or sample limitation.

Choose your support and challenge

- Supported: Match four after-wipe picture cards to Codes A–D and complete one two-code comparison.
- Standard: Complete all four rows, keep one repeat/check note and write a conclusion comparing at least two codes.
- Stretch: Evaluate the relative pattern, explain an unexpected result and propose one safe improvement without claiming a universal hardness value.

Show what you know

Use two code results to give a relative-hardness conclusion and state one limit.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.